

Spectral Admissibility in the Q_8 Cayley Graph: A Minimal Toy Model for Bounded-Flux Mode Selection

Working note – Cosmochrony programme

March 7, 2026

Abstract

We establish the first analytically tractable instance of the spectral admissibility programme outlined in the Cosmochrony framework. On the undirected Cayley graph of the quaternion group Q_8 , with generators $S = \{\pm i, \pm j, \pm k\}$, the local bounded-flux constraint $|\partial_t \chi_v| \leq c_\chi$ induces a hierarchy of maximal admissible effective amplitudes across irreducible spectral sectors. In the linearised dispersion regime, the non-abelian irreducible sector admits a maximal effective amplitude larger than the nontrivial abelian sectors by a factor of $\sqrt{4/3} \approx 1.155$. This constitutes a minimal model demonstration that bounded substrate capacity may structurally favour non-abelian relational modes, independently of any dynamical postulate.

1 Setting and Motivation

The Cosmochrony framework identifies the physical spectrum with the residue of two successive constraints acting on the relational operator $\mathcal{L}_\chi$:

1. **Topological discretisation:** the fibre structure of the non-injective projection Π organises modes into discrete winding sectors.
2. **Bounded-flux filtering:** the saturation bound $|\partial_t \chi| \leq c_\chi$ eliminates modes whose local evolution would exceed the substrate's relational capacity.

The goal of this note is to make step 2 explicit and calculable on the simplest non-trivial finite graph encoding a quaternionic algebraic structure, as a minimal discrete proxy for $SU(2)$ -type sectors.

2 The Q_8 Cayley Graph

Let $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ denote the quaternion group of order 8. We define the undirected Cayley graph $\Gamma(Q_8, S)$ with generating set

$$S = \{\pm i, \pm j, \pm k\}. \tag{1}$$

Since S is closed under inversion ($s^{-1} \in S$ for all $s \in S$) and does not contain the identity, $\Gamma(Q_8, S)$ is a well-defined, connected, undirected, 6-regular graph on 8 vertices.

Remark 1. *The choice $S = \{i, j, k\}$ (without inverses) would yield a directed graph with an asymmetric adjacency operator. For a self-adjoint Laplacian, the full symmetric set $S = \{\pm i, \pm j, \pm k\}$ is required. The resulting graph is 6-regular, not 3-regular.*

2.1 Adjacency Operator and Graph Laplacian

The adjacency operator A of $\Gamma(Q_8, S)$ acts on functions $f: Q_8 \rightarrow \mathbb{C}$ by

$$(Af)(g) = \sum_{s \in S} f(gs) = \sum_{s \in \{\pm i, \pm j, \pm k\}} f(gs). \quad (2)$$

The combinatorial graph Laplacian is

$$\mathcal{L} = dI - A, \quad d = |S| = 6, \quad (3)$$

where I is the identity on $\ell^2(Q_8)$. The eigenvalues of $\mathcal{L}$ are $\lambda = 6 - \mu$, where μ ranges over the eigenvalues of A .

3 Spectral Decomposition via Representation Theory

Since Q_8 is a finite group, $\ell^2(Q_8) \cong \mathbb{C}[Q_8]$ decomposes as a Q_8 -module into irreducible representations. By the Peter–Weyl theorem for finite groups:

$$\mathbb{C}[Q_8] \cong \bigoplus_{\rho \in \widehat{Q_8}} V_{\rho}^{\oplus \dim \rho}, \quad (4)$$

where the sum runs over the five irreducible representations of Q_8 .

3.1 Irreducible Representations of Q_8

Q_8 has five irreducible representations:

Label	Dimension	$\rho(i)$	$\rho(j)$	$\rho(k)$
ρ_0 (trivial)	1	1	1	1
ρ_1	1	1	-1	-1
ρ_2	1	-1	1	-1
ρ_3	1	-1	-1	1
ρ_4 (quaternionic)	2	$\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$

The dimension check: $1^2 + 1^2 + 1^2 + 1^2 + 2^2 = 8 = |Q_8|$. ✓

3.2 Eigenvalues of the Adjacency Operator

For any irreducible representation ρ , the adjacency operator acts as the scalar $\mu_{\rho} \cdot I_{V_{\rho}}$ on V_{ρ} . Since S is a union of conjugacy classes of Q_8 , the corresponding element of the group algebra is central, and Schur's lemma gives

$$\mu_{\rho} = \frac{1}{\dim \rho} \sum_{s \in S} \chi_{\rho}(s), \quad (5)$$

where χ_{ρ} is the character of ρ .

1D sectors. For the trivial representation ρ_0 : $\chi_{\rho_0}(s) = 1$ for all $s \in S$, so $\sum_{s \in S} \chi_{\rho_0}(s) = 6$ and

$$\mu_{\rho_0} = \frac{1}{1} \cdot 6 = 6, \quad \lambda_{\rho_0} = 6 - 6 = 0. \quad (6)$$

For ρ_1 : $\chi_{\rho_1}(\pm i) = 1$ and $\chi_{\rho_1}(\pm j) = \chi_{\rho_1}(\pm k) = -1$, so $\sum_{s \in S} \chi_{\rho_1}(s) = 2(1) + 4(-1) = -2$, giving $\mu_{\rho_1} = -2$ and $\lambda_{\rho_1} = 8$. By the symmetry of Q_8 under permutations of $\{i, j, k\}$, $\mu_{\rho_2} = \mu_{\rho_3} = -2$ and $\lambda_{\rho_2} = \lambda_{\rho_3} = 8$.

2D sector. For the quaternionic representation ρ_4 , we compute the character at each generator. The matrices $\rho_4(i)$, $\rho_4(-i)$, $\rho_4(j)$, $\rho_4(-j)$, $\rho_4(k)$, $\rho_4(-k)$ each have trace zero: for instance, $\text{tr}(\rho_4(i)) = \text{tr}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = 0$, and $\text{tr}(\rho_4(-i)) = 0$; similarly for j and k . Thus $\sum_{s \in S} \chi_{\rho_4}(s) = 0$ and

$$\mu_{\rho_4} = \frac{1}{2} \cdot 0 = 0, \quad \lambda_{\rho_4} = 6 - 0 = 6. \quad (7)$$

3.3 Summary of the Spectrum

Sector	λ_n	Multiplicity	Type
ρ_0	0	1	Trivial (zero mode)
ρ_1, ρ_2, ρ_3	8	3	Nontrivial abelian
ρ_4	6	4	Non-abelian (quaternionic)

Total dimension: $1 + 3 + 4 = 8$. ✓

4 Bounded-Flux Admissibility Condition

4.1 Setup

Let the substrate field $\chi: Q_8 \times \mathbb{R} \rightarrow \mathbb{R}$ evolve on the graph, subject to the local bounded-flux constraint

$$|\partial_t \chi_v(t)| \leq c_\chi \quad \forall v \in Q_8, \forall t. \quad (8)$$

Consider a single-mode excitation

$$\chi_v(t) = a_n \psi_n(v) \cos(\omega_n t), \quad (9)$$

where ψ_n is an eigenmode of $\mathcal{L}$ with eigenvalue λ_n , normalised in $\ell^2(Q_8)$, and $a_n > 0$ is the mode amplitude. The time derivative at vertex v is

$$\partial_t \chi_v(t) = -a_n \omega_n \psi_n(v) \sin(\omega_n t). \quad (10)$$

The constraint (??) requires

$$|\omega_n| \cdot a_n \cdot |\psi_n(v)| \leq c_\chi \quad \forall v \in Q_8. \quad (11)$$

The binding constraint is at the vertex where $|\psi_n(v)|$ is maximal:

$$|\omega_n| \cdot a_n \cdot \|\psi_n\|_\infty \leq c_\chi. \quad (12)$$

4.2 Effective Amplitude and Admissibility Envelope

Definition 1. The effective amplitude of a mode (ψ_n, a_n) is

$$A_n := a_n \|\psi_n\|_\infty. \quad (13)$$

A mode is dynamically admissible (in the linearised regime) if

$$|\omega_n| \cdot A_n \leq c_\chi. \quad (14)$$

In the linearised dispersion approximation, $\omega_n^2 \approx \lambda_n$, so the maximal admissible effective amplitude is

$$A_n^{\max} = \frac{c_\chi}{\sqrt{\lambda_n}}. \quad (15)$$

This defines the *admissibility envelope* in the (λ_n, A_n) plane.

Remark 2. The linearised dispersion $\omega_n^2 = \lambda_n$ is valid for $A_n \ll c_\chi / \sqrt{\lambda_n}$. The non-linear (Born-Infeld) correction modifies this relation near saturation and requires a modal reduction of the DBI action; it is deferred to a separate calculation. The envelope (??) should therefore be understood as the admissibility boundary in the weakly non-linear regime.

5 Main Result

Proposition 1 (Spectral admissibility hierarchy on Q_8). *In the undirected Cayley graph $\Gamma(Q_8, \{\pm i, \pm j, \pm k\})$, under the local bounded-flux constraint (??) and linearised dispersion, the maximal admissible effective amplitude satisfies*

$$A_{\rho_4}^{\max} = \frac{c_\chi}{\sqrt{6}} > \frac{c_\chi}{\sqrt{8}} = A_{\rho_k}^{\max}, \quad k = 1, 2, 3. \quad (16)$$

The non-abelian sector ρ_4 occupies a strictly larger admissible region in the space of effective mode amplitudes than any nontrivial abelian sector, by a factor

$$\frac{A_{\rho_4}^{\max}}{A_{\rho_k}^{\max}} = \sqrt{\frac{8}{6}} = \sqrt{\frac{4}{3}} \approx 1.155. \quad (17)$$

Proof. Direct substitution of $\lambda_{\rho_4} = 6$ and $\lambda_{\rho_k} = 8$ into the admissibility envelope (??). $\square$

Remark 3. *The trivial sector ρ_0 has $\lambda = 0$ and thus formally unbounded $A^{\max}$. It corresponds to the uniform zero mode of the connected graph. In the present analysis it is factored out, since we are interested only in non-trivial excitations above the connected background.*

6 Interpretation

Proposition ?? establishes that, at fixed substrate capacity c_χ , the non-abelian irreducible sector of the quaternion Cayley graph sustains a larger local effective amplitude before violating the bounded-flux constraint. In the language of the Cosmochrony programme:

- The admissibility envelope $A_n^{\max} = c_\chi / \sqrt{\lambda_n}$ defines a *geometry of exclusion* in the space of relational modes: modes above the envelope are dynamically inadmissible.
- The non-abelian sector ρ_4 lies strictly below this envelope for a larger range of effective amplitudes than the nontrivial abelian sectors ρ_1, ρ_2, ρ_3 : the bounded-flux constraint is less restrictive for non-abelian modes, enlarging their admissible window by a factor of $\sqrt{4/3}$.
- This provides a minimal admissibility advantage for non-abelian relational modes, without any dynamical postulate beyond the local flux bound.

Remark 4 (Real field and complex eigenmodes). *The substrate field χ is taken real-valued. The natural eigenmodes of the 2D sector are matrix coefficients of ρ_4 , which are complex-valued. One may either work in $\ell^2(Q_8, \mathbb{C})$ and take real and imaginary parts to obtain real eigenmodes, or directly choose a real basis for each eigenspace. The admissibility bound on A_n is basis-independent (it depends on $\|\psi_n\|_\infty$, which may vary within a degenerate eigenspace); this basis dependence is addressed in Step 1 of Section ??.*

This result does not yet establish:

- that the non-abelian sector will dominate in a full dynamical simulation;
- the origin of the eigenvalue gap $\lambda_{\rho_4} = 6 < 8 = \lambda_{\rho_k}$ from the topological structure of Π alone;
- the Born-Infeld correction to the admissibility envelope near saturation.

These remain open and constitute the next steps of the programme.

7 Next Steps

The natural continuation of this calculation proceeds as follows.

Step 1: explicit eigenmodes and $\|\psi_n\|_\infty$. Choose an explicit ℓ^2 -normalised basis in each eigenspace using the matrix coefficients of the representations. For each basis vector, compute $\|\psi_n\|_\infty$ explicitly. In degenerate eigenspaces (multiplicity > 1 , in particular the 2D sector with multiplicity 4), determine the basis dependence of the admissibility bound $a_n^{\max} = c_\chi/(\sqrt{\lambda_n} \|\psi_n\|_\infty)$ and identify the basis minimising or maximising $\|\psi_n\|_\infty$ within the eigenspace. Expected values: $\|\psi_n\|_\infty = 1/\sqrt{8}$ for 1D sectors and $\|\psi_n\|_\infty \leq 1/\sqrt{2}$ for the 2D sector (upper bound from L^2 -normalisation).

Step 2: modal DBI reduction. For a single mode $\chi_v(t) = q(t)\psi_n(v)$, substitute into the discrete DBI Lagrangian

$$L = -c_\chi^2 \sum_v \sqrt{1 - \dot{\chi}_v^2/c_\chi^2} - \frac{1}{2} \sum_{v,w} \chi_v \mathcal{L}_{vw} \chi_w, \quad (18)$$

and derive the effective equation for $q(t)$. Extract the amplitude-dependent frequency $\omega_n(a_n)$ via harmonic balance. This yields the corrected admissibility envelope replacing (??).

Step 3: dynamical simulation. Simulate the full Q_8 substrate with the DBI constraint imposed locally at each vertex. Track the modal decomposition $A_n(t) = a_n(t)\|\psi_n\|_\infty$ over time. Test whether modes initialised above the envelope (??) decay, and whether modes in the non-abelian sector survive longer than those in nontrivial abelian sectors at comparable initial A_n .

Step 4: extension to larger graphs. Replace Q_8 by the Cayley graph of a group approximating S^3 more faithfully at large N (e.g. the binary icosahedral group $2I$ of order 120, or $\text{SL}(2, \mathbb{F}_p)$ for small primes p). Test whether the spectral admissibility hierarchy persists under this refinement. A particularly natural continuation is provided by Cayley graphs in Ramanujan families – such as the Lubotzky–Phillips–Sarnak construction on $\text{SL}(2, \mathbb{F}_p)$ – where the spectral gap is asymptotically optimal. This would allow the admissibility hierarchy to be tested in graph sequences with extremal expansion properties, directly connecting bounded-flux filtering to the theory of expander graphs.